

Performance Test

Test Run ID: c0554b59-73c6-4185-968d-4f79fe9de73e

This chapter provides a summary of the test run metrics. The tables contain the aggregated values of the metrics for the entire test run.

Date	05 June 2026	Max VUs Tested	80
Executor	constant-vus	Tool	k6

1. Executive Summary

The target endpoint for this performance test is the "https://quickpizza.grafana.com" URL. The testing type is a constant load scenario, where 80 Virtual Users (VUs) are maintained throughout the 30-second test duration. The load profile consists of a constant arrival rate of 80 VUs, with a steady state of 80 VUs and no ramp-up or ramp-down stages.

Metric	Value
Minimum TPS	55.92
Average TPS	61.04
Maximum TPS	126.75

2. Performance Summary

Trends

metric	avg	max	med	min	p90	p95	p99
http_req_duration	234.57ms	747.75ms	233.38ms	217.23ms	241.98ms	247.87ms	-
http_req_waiting	233.73ms	519.83ms	232.96ms	217.05ms	241.39ms	247.48ms	-
http_req_connecting	22.63ms	276.94ms	0.00ms	0.00ms	0.00ms	236.14ms	-
iteration_duration	1283.54ms	2220.81ms	1235.27ms	1218.40ms	1271.79ms	1722.74ms	-

Counters

metric	count	rate
data_received	6.86 MB	224.84 kB/s
data_sent	402.88 kB	12.90 kB/s
http_reqs	1,906	61.04/s
iterations	1,906	61.04/s

Rates

metric	rate
checks_succeeded	100.00% (1906/1906)
checks_failed	0.00% (0/1906)
http_req_failed	0.00% (0/1906)

Gauges

metric	value
vus	15 (min: 15, max: 80)
vus_max	80 (min: 80, max: 80)

Checks

Check Name	Passed	Failed	Success Rate	Status
status is 200	1906	0	100.00%	Pass

3. Advanced Analysis

3.1 Error Analysis

- **Network/Connection errors:** None detected.
- **Application errors:** None detected.
- **Infrastructure errors:** None detected.

- **Warnings:** A single warning was logged due to unknown fields in the options exported in the script: ``json: unknown field "executor"``.

3.2 Observation

The performance test results indicate a stable and consistent system behavior. However, upon closer inspection of the logs and metrics, we can observe some subtle patterns that warrant further investigation.

- **Client timeouts:** The ``http_req_duration`` metric shows an average duration of 234.57ms, with a maximum duration of 747.75ms. This suggests that some requests are taking significantly longer than others, potentially due to client timeouts.
- **503 Service Unavailable:** Although no explicit 503 errors were detected, the ``http_req_duration`` metric shows a high maximum duration, which could indicate that the system is experiencing upstream resets or gateway timeouts.
- **Session not found:** The ``status is 200`` check passed 1906 times, but there is no explicit check for session not found errors. This could indicate that the system is not properly handling session initialization or persistence.
- **Silent 500 session initialization failures:** The lack of explicit 500 errors in the logs suggests that session initialization failures may be occurring silently, without any visible errors or warnings.

3.3 Final Recommendations

To address the identified bottlenecks and errors, we recommend the following:

1. **Advanced Connection Management:** Implement a connection pooling mechanism to reduce the overhead of establishing new connections for each request.
2. **Session Persistence sticky affinity/Redis cache:** Implement a session persistence mechanism to ensure that sessions are properly stored and retrieved, reducing the likelihood of session not found errors.
3. **Gateway resilience Istio outlier circuit breaker:** Implement an outlier detection mechanism to detect and prevent upstream resets or gateway timeouts.
4. **Observability tracing and standardized log levels:** Implement a tracing mechanism to provide detailed insights into system behavior and log levels to ensure that all errors and warnings are properly logged and visible.
5. **Database read/write splitting and PgBouncer connection pooling:** Implement a read/write splitting mechanism to reduce the load on the database and implement a connection pooling mechanism to reduce the overhead of establishing new connections for each request.

3.4 Conclusion

Based on the performance test results and analysis, we conclude that the system is stable and consistent, but with some subtle patterns that warrant further investigation. The recommended changes will help to address the identified bottlenecks and errors, ensuring that the system is properly optimized for performance and reliability.